

WSI Viewer Quick Guide

Viewing, Multi-view, Hierarchy, channel viewer, annotations, and export

1. Open and navigate

- Search or filter cases in the upper left, then choose a slide from the list.
- Drag to pan; scroll or pinch to zoom. Home resets to the fitted view.
- Each open pane shows its own scale bar when the slide is calibrated.
- Press **H** to hide or show the left image browser. Its tabs are **Slides**, **Image**, **Annotations**, and **Hierarchy**.
- Click **?** for the keyboard shortcuts legend; Shift-click **?** for the Help directory.

2. Adjust channels

- Press **C** or use Brightness & Contrast. Changes affect display only, not source pixels.
- Use checkboxes to show or hide channels; set LUT color, min/max, gamma, and opacity. An empty colored outline means the channel is off; a × inside the outline means it is on. The box is never filled with the channel color.
- Drag the white min/max lines on the histogram, or the Channel min/max sliders, to set the display window. Double-click **Channel min** or **Channel max** to type a value.
- Use **Reset display** or **Recompute auto**.
- **View** ▶ **Show channel viewer** opens a detached window of each visible channel plus Composite. It follows the cursor by default and stays in register with the main window while you zoom.
- Right-click the channel viewer to sync to cursor or viewer center, set zoom relative to the main window, or show all channels.

3. Multi-view

- Right-click the image (not an annotation) or use the **View** menu. The tree matches QuPath: **Multi-view**, **Cell display**, **Display** (Show analysis, Brightness/Contrast, overlay 400%–1%), and **Set tool**. Nested lists fly left or right from the pointer so only one branch is open.
- Set grid size (1×1, 1×2, 2×1, 2×2, 3×3), or add/remove a row or column. Close images in a row or column before removing it.
- Click a pane to make it active (red outline), then choose a slide or drag a slide from the list onto that pane.
- A newly opened image fits the pane it occupies. Changing the grid re-fits open images to their new cells.
- **Synchronize viewers** (**Ctrl+Shift+S** / ⌘⇧S / **Ctrl+Alt+S**) turns on automatically when two or more panes have images. **Match viewer resolutions** matches magnification using pixel size.
- **Close viewer** clears the active pane. **Detach** / **Attach** moves a pane into its own window when the grid has more than one cell.

4. Classify

- Use the **Classify** menu (same tree as QuPath).
- **Object classification** trains, saves, loads, and applies classifiers to detections. Draw classified annotations over detections, or right-click an annotation and choose **Set class**. Shortcut **Ctrl+Shift+D** / ⌘⇧D.
- **Pixel classification** trains, saves, loads, and applies classifiers to annotations, or creates a thresholder. Shortcut **Ctrl+Shift+P** / ⌘⇧P.
- **Training images** creates region annotations, stores a training-image record, duplicates regions per channel, and splits the catalog into train/validation/test.

5. Work with annotations

- Use **Annotations (A)** and **Names (N)** to show or hide overlays. **D** toggles automated cell detections; **F** and **Shift+F** toggle detection and annotation fills.
- **View** ▶ **Cell display** switches detections among nuclei & cell boundaries, nuclei only, cell boundaries only, and cell centroids only. In AI Labs, each detector (StarDist, Cellpose, QuPath Cell Detection) has a cell-expansion method and amount so the cytoplasm ring can be scaled or offset from the nucleus. **Selected annotation** limits detections to the annotation outline (not its bounding box). Choose whether nuclei that cross that outline are excluded, included in full, or truncated to the border. For fluorescence StarDist, leave **Nuclear channel (recommended)** so only the nuclear stain is used. After Quantify or color-coding, **Heat Map** shows those measurement colors on the detections.
- Select one annotation to view or edit its name. Enter or leave the name field to commit; Escape cancels a draft.
- Tool keys: **M** move, **S** select, **R** rectangle, **O** ellipse, **L** line, **P** polygon, **V** polyline, **B** brush, **W** wand, . points, **Z** zoom. Enter seals a polygon/polyline/wand.
- Click-drag a **locked** annotation to pan the image; annotations and detections stay in register. Unlocked annotations still move.
- Select one, or Shift-click several, then press Delete/Backspace, use **Delete**, or right-click the annotation for Lock/Delete or **Set class**. If the annotation has detections or other objects inside it, you are asked whether to keep those descendants: **Yes** keeps them, **No** deletes them too, **Cancel** leaves everything. Deletion cannot be undone.
- The **Hierarchy** tab lists the image, annotations, and detections as a tree. Expand or collapse branches, lock the image or an annotation, and inspect the selected object under **Measurements** or **Description**. Filter the table by key.

6. Protect recent annotation work

- Saving is asynchronous and currently has no visible Saved indicator.
- After creating, moving, renaming, or deleting, pause several seconds before switching images, refreshing, or closing.
- For important work, return to the image to confirm persistence.

7. Export, Z-stack, and overview

- **Selected annotation** exports the selected region. **Entire View** means the visible region, not necessarily the whole slide.
- Native exports are limited to **16,000,000 pixels**. Zoom in and retry if no file appears.
- Z-stacks: **View** ▶ **Open Z-stack controller** (or image right-click) reopens the focal-plane window. **Alt+wheel** or arrow keys change focal plane. Slide overview (**Ovw**) opens a thumbnail navigator.
- Double-click the magnification readout to type an exact value.

8. Environment and safety

- Production normally has no warning banner.
- **DEVELOPMENT** or **STAGING** means validation only and not clinical use.
- Use only images appropriate for the assigned environment.
- Never copy sensitive production material into development or staging.

9. Help and feedback

- Click **?** for every keyboard shortcut currently bound in the viewer.
- Shift-click **?** for the Help directory (User Guide, this Quick Guide, Admin & Ops).
- **Ctrl+Shift+F** opens Pilot Feedback. Do not paste clinical image data into feedback.

Report: blank/distorted images, missing or changed annotations, failed exports, login errors, or unusual delays. Report the image name and approximate time; do not send sensitive image data outside the approved support process.

Before reporting a display problem, try one normal refresh - but never refresh while a recent annotation change may still be saving.